Nguyen Hoang Nguyen

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Experience

AI Engineer *Ho Chi Minh City, Vietnam*
FPT Software AI Center *Dec 2023 – Present*

- Built a RAG indexing pipeline on Apache Airflow processing 500+ multi-format documents, using VLMs and TableFormer to extract nested tables and diagrams from multilingual PDFs, Excel, and PPTX, reaching **80% context recall**.
- Researched and shipped agentic systems on LangGraph – dynamic tool calling, planning, and multi-step reasoning for autonomous query resolution, plus a document-generation service that cut manual documentation effort by **22%**.
- Automated legacy-code understanding across 1M+ lines and 7 languages on 4 mainframe systems: a Gzip-compression + k-NN classifier at **90% accuracy**, and custom ANTLR4 parsers reaching **88% COBOL syntax coverage**.

Skills

3D Reconstruction & Novel View Synthesis: 3D Gaussian Splatting, gsplat, Neural Harmonic Textures, 3DGUT, COLMAP structure-from-motion, camera calibration, MCMC densification, PSNR/SSIM/LPIPS evaluation, Rerun

Simulation & Robotics: NVIDIA Isaac Sim, NVIDIA Isaac Lab, Genesis, Reinforcement Learning, PPO, Reward Shaping, MDP Design, skrl

ML Engineering & LLMs: Python, PyTorch, OpenCV, NumPy, VLMs, LoRA & QLoRA, LangChain, LangGraph, Agentic RAG, Weights & Biases, pytest

Infrastructure: Modal, Docker, Linux, Git, FastAPI, Azure, Apache Airflow, PostgreSQL, MongoDB, Milvus, Neo4j, Redis, Scikit-learn, ANTLR4

Projects

Saigon Digital Twin: Photorealistic 3D City Streets from Public Imagery *(Personal R&D, 2026)*

- Building a walkable, photorealistic 3D model of downtown Ho Chi Minh City from imagery that already exists, **fusing 21 dates of satellite photos with 100+ street-level 360° panoramas** instead of commissioning a costly survey; the result can be explored in a web browser.
- Designed the end-to-end pipeline, from camera alignment to **Gaussian Splatting** training on cloud GPUs (Modal), with AI models that remove cars, pedestrians, shadows and sky so only the permanent city is kept.
- Ran **30+ tracked experiments** under a self-imposed **$200 compute budget**, with a fixed quality scorecard and regression checks so each improvement is measured on views the model never saw.

3D Reconstruction of Drone-Captured Tower Sites, Reflective and See-Through Scenes *(VAI Novel-View-Synthesis Competition)*

- Reconstructed camera poses with **COLMAP** structure-from-motion, re-undistorted the input images with **Lanczos4** resampling, then trained **Neural Harmonic Textures** on **gsplat** through a **3DGUT** rasterizer, growing to 9M primitives with **MCMC densification** under a combined **L1 + D-SSIM + LPIPS** loss.
- Built an end-to-end pipeline turning photo sets into photorealistic, measurable 3D models, focusing on scenes where conventional reconstruction breaks down: a **thin lattice telecom tower** shot by drone over a crowded city block, which leaves almost no solid structure to lock onto; a **bonsai on a reflective glass table**, where the reflection reads as real geometry; and a **chair with a see-through woven mesh back**, achieving **PSNR 25.76, SSIM 0.852 and LPIPS 0.134** on independent novel-view evaluation.

Code as Policy *(Genesis, Franka Panda)*

- Used an **LLM as the robot's high-level controller**: plain-language instructions are converted into Python commands that drive a Franka Panda arm in the **Genesis** physics simulator, guided by an overhead

depth camera with **open-vocabulary object detection (YOLOE)** that locates each object on the table in 3D.

- Designed the **motion planning** for collision-free execution: **RRTConnect** plans the approach, and every grasp is validated before the arm moves.

Reinforcement Learning for Robotic Grasping and Lifting (UR10 + Robotiq 2F-140)

- Built and trained a PPO policy in NVIDIA Isaac Lab to grasp a cube and lift it to a randomized target pose with a UR10 arm and Robotiq 2F-140 parallel gripper, achieving **100% grasp rate** and **93% target-reach success** within 1 cm of randomized targets.
- Designed the MDP: joint-position + binary-gripper action space, observation space covering joint state and object/target poses, and a multi-term reward covering reach, lift, and goal tracking, regularized by smoothness and joint-behavior penalties.
- Reduced arm vibration at the target pose via a goal-proximity velocity penalty and tuned PD gains, cutting the action-rate penalty by **38%** without affecting task success.

Vietnamese-English Translation with Teencode and Slang Understanding

- Fine-tuned Vistral-7B (4-bit, LoRA) on a curated 9-task instruction dataset, achieving **34.9 BLEU** on VLSP2020 and beating GPT-3.5 and Google Translate on three Vietnamese social-media benchmarks by up to **+8.1 BLEU**. Written up for publication.

Education

Ton Duc Thang University — *Ho Chi Minh City, Vietnam*
Bachelor of Science in Software Engineering — *Sep 2020 – Aug 2024*

- GPA 3.44/4.0. Undergraduate program taught in English.
- Member of the university's ICPC training team; core member of ICON Club (technical blogs, seminars, full-stack quiz platform).

Feng Chia University – Exchange Student, GPA 3.5/4.0 — *Taichung, Taiwan*, *Jul 2023 – Aug 2023*

Publications

Enhancing Vietnamese to English Translation with Teencode and Slang Understanding — Under Review

Nguyen Hoang Nguyen, Phuoc Tran

EAI International Conference on Artificial Intelligence for Communications and Networks

Honors, Awards and Certificates

University Study Scholarship, 2020–2023 (academic performance) — *2023*

3rd place, Academic contest "Recursion" 2022 — *2022*

University Entrance Scholarship (national graduation exam scores) — *2020*

NLP Specialization; Generative AI with LLMs *(DeepLearning.AI)* — *2023*